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EEG dataset of consumer- and research-grade systems

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An open electroencephalography (EEG) dataset is presented for evaluating consumer-grade EEG devices in comparison with a research-grade EEG device. Datasets were collected from 30 participants using four consumer-grade EEG devices (BrainLink Pro, NeuroNicle FX2, MindWave Mobile 2, Muse 2), including two single-channel headsets (BrainLink Pro and MindWave Mobile 2), and one research-grade EEG device (DSI-24). Each participant completed four experimental paradigms: eye blinks, jaw clenching, head movements with eyes open, and head movements with eyes closed, with the resting-state EEG data recorded before and after each task. The dataset enables the validation of device signal quality, the assessment of neural features such as alpha-band activity, and the examination of robustness to movement-induced artifacts. The dataset is publicly available on Figshare.

Background & Summary

Electroencephalography (EEG) has traditionally relied on costly equipment and specialized expertise[1]. However, with growing interest in brain–computer interfaces (BCIs), which directly connect the human brain to computer systems for interpreting neural signals and enabling communication, the market for consumer-grade electroencephalography devices[2] has expanded, making EEG measurements more accessible to the public.

However, consumer-grade EEG devices have a relatively short development history compared with research-grade systems. Their simplified designs often lack rigorous validation for signal quality and usability, although several studies have begun to assess their performance. Specifically, some investigations examined whether canonical EEG features, such as event-related potentials (ERPs) and quantitative EEG (QEEG) measures, could be reliably detected during the cognitive paradigms commonly employed in BCI research, including the oddball task, n-back task, and stop-signal task[3,4]. Other studies assessed the reliability of EEG characteristics observed during resting-state recordings to determine device performance[4,5]. Additional work evaluated the detectability of high-amplitude artifacts, such as eye blinks[6,7], as well as potential signal delays, distortions[3], and overall usability[8]. Despite prior efforts, a standardized framework for evaluating consumer-grade EEG devices remains absent. Only a limited number of public datasets are explicitly designed for device-oriented evaluation, and these resources typically focus on specific conditions (e.g., wet–dry electrode comparisons within a single paradigm in wearable SSVEP)[9]. In addition, some artifact-focused datasets are released primarily as denoising benchmarks (e.g., EEGdenoiseNet) or as large clinical EEG corpora (e.g., TUH-EEG), rather than for staged robustness evaluation of consumer-grade wearables under a unified protocol[10]. To address this gap, we introduce a top-down, three-level evaluation model progressing from artifact detectability to characteristic EEG features and movement sensitivity, under which the present dataset was collected.

This dataset comprises EEG recordings from 30 participants, including sessions with eye blinks, jaw clenching, and head movements, as well as resting-state EEGs. Such recordings enable the systematic evaluation of EEG device signal quality and provide opportunities for examining how movement-related artifacts affect EEG measurements. By making this dataset publicly available, we aim to support further research on the validation of consumer- and research-grade EEG systems, as well as to facilitate the analyses of artifact-related changes in EEG signals across diverse applications.

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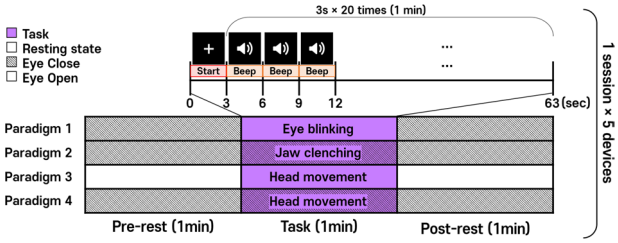


Fig. 1 Experimental paradigm.

Methods

Ethics statement. Participants were recruited via an IRB-approved advertisement posted on university-, department-, and laboratory-level bulletin boards and through social media channels. To minimize potential undue influence and ensure voluntary participation, students affiliated with the principal investigator's department were not eligible to participate. All participants provided written informed consent for study participation and for the public sharing of de-identified EEG recordings via an open repository. Direct identifiers (e.g., name and contact information) were collected only for administrative purposes and were stored separately from the EEG recordings with restricted access. The publicly released dataset contains no direct identifiers; participants are referenced only by randomly assigned subject codes (e.g., sub-01), and only aggregate demographic information (e.g., group means) is reported in the manuscript.

Participants and environment. A total of 30 healthy adults (16 females and 14 males; age range 19–27 years; mean age 23.2 years) participated in this study. All participants provided written informed consent before participation. The Institutional Review Board of Handong Global University (IRB No. 2023-HGUR026) approved the study protocol, and all procedures were conducted in accordance with relevant guidelines and regulations.

All participants were tested under identical experimental conditions, including the same desk, chair, and monitor setup. They were instructed to place their hands on a designated spot on the desk and minimize movement during the recordings.

Experimental paradigm. The experimental paradigm, which followed a top-down structure, was designed to technically assess whether consumer-grade EEG devices can capture expected EEG signal characteristics under simple and controlled experimental conditions. First, we assessed whether large-amplitude artifacts, such as eye blinks, could be reliably detected. Second, we examined whether the recorded signals exhibited characteristic EEG features. Finally, we evaluated the movement sensitivity of consumer-grade EEG devices intended for use across a range of scenarios.

To validate the framework, the experiment comprised four paradigms. Each paradigm followed the same structure: one minute of resting-state EEG before task execution, a one-minute task period consisting of 20 repetitions at three-second intervals, and one minute of resting-state EEG after the task (Fig. 1).

The first paradigm involved an eye-blink task. Participants kept their eyes closed during the initial resting-state recording and opened their eyes at the onset of a beep signal. They then performed 20 eye blinks, each synchronized with a beep presented every three seconds. During the post-task resting-state recording, participants again kept their eyes closed.

The second paradigm involved a jaw-clenching task. Following a one-minute resting-state recording, participants performed 20 jaw clenches, each cued by a beep at three-second intervals. This paradigm was conducted with participants keeping their eyes closed throughout the session.

The third paradigm involved head movements with eyes open. After a one-minute pre-task resting-state recording, participants performed 20 head turns guided by auditory beep cues. A main beep was presented every 3 s, and participants alternated the head-turn direction at each main beep (left-to-right, then right-to-left). To help participants maintain a consistent movement speed and avoid moving too quickly or too slowly, an additional sub-beep was presented every 1 s. This movement sequence was repeated 20 times (approximately 1 min of active head movement), followed by a one-minute post-task resting-state recording.

The fourth paradigm was identical to the third, except that these participants kept their eyes closed during the entire session.

Timing and scheduling of recording sessions. For each participant, all recordings were completed within a single day. Participants wore the five EEG devices sequentially, and for each device, they completed the same four paradigms. The session order was designed such that the research-grade device (DSI-24) was always placed third (i.e., in the middle of the sequence), while the order of the remaining consumer-grade devices was randomized and counterbalanced across participants to achieve an approximately uniform distribution of device orders.

Event markers were recorded to delimit the experimental phases and task repetitions, and the same integer event codes (1–5; Table 3) were provided in both the EDF and MAT files, with code 3 emitted at the onset of each task repetition.

In addition to EEG recordings, participants completed questionnaires administered at three time points: before the experiment, after each recording session with the device, and after completing all sessions. The questionnaires were presented in Korean (native language). The translated items are provided in Table 1, and the survey was based on the System Usability Scale (SUS)[11].

Question time	Items	Remarks
Before the experiment	How would you describe your current physical condition? Please evaluate your status based on the past month. (0~10)	0: Very poor 10: Very nice
	How would you describe your current mental condition? Please evaluate your status based on the past month. (0~10)	0: Very poor 10: Very nice
After each session	What is the maximum duration you can wear this device (min.)?	10, 20, 30, 60, more than 60 min.
	How uncomfortable is this device to wear? (1–10)	1: Very uncomfortable 10: Not uncomfortable at all
	Would you like to wear this device again? (1–10)	1: I would not want to wear it again at all. 10: I would very much like to wear it again.
	Is this device's design aesthetically pleasing? (1–10)	1: Not aesthetically pleasing at all 10: Very aesthetically pleasing
	Is this device easy to wear? (1–10)	1: Very difficult to wear 10: Very easy to wear
	Do you think most people would easily get accustomed to using this device? (1–10)	1: Not likely to get accustomed at all 10: Very likely to get accustomed
After the experiment	Please rank Devices A, B, C, D, and E in order of difficulty to wear.	
	Please rank Devices A, B, C, D, and E in order of discomfort when worn.	
	Which device do you prefer the most?	

Table 1. Questionnaire items based on the System Usability Scale (SUS).

Test devices	Sampling rate	The number of channels	Recording software
Brain Link Pro	512 Hz	1	OpenViBE 2.2.0
FX2	250 Hz	2	NeuroNicle FX2 viewer
Mindwave2	512 Hz	1	OpenViBE 2.2.0
Muse2	256 Hz	4	BlueMuse and OpenViBE 2.2.0
DSI-24	300 Hz	21	OpenViBE 2.2.0

Table 2. Test devices information.

Data collection. EEG data were recorded using four consumer-grade devices and one research-grade device for comparison. Device information is provided in Table 2.

BrainLink Pro (BLP) of Macrotellect Inc., released in 2018, is a single-channel wireless EEG device featuring a dry electrode at Fp1 and a reference on the left ear. It supports a maximum sampling rate of 512 Hz. Signal acquisition in this study was conducted using OpenViBE.

The NeuroNicle FX2 (FX2) by Laxtha Inc. is a wireless dry-electrode EEG device featuring two electrodes over the left and right frontal regions (approx. Fp1 and Fp2) and a reference on the left ear. It supports a maximum sampling rate of 512 Hz. Signal acquisition was performed using the proprietary NeuroNicle FX2 Viewer software.

The MindWave Mobile 2 (MW2) by NeuroSky, released in 2018, is a single-channel wireless EEG device featuring a dry electrode at Fp1 and a reference on the left ear. It supports a maximum sampling rate of 512 Hz. Signal acquisition in this study was conducted using OpenViBE.

The Muse 2 (InteraXon Inc., 2018) is a wireless dry-electrode EEG device featuring four electrodes (AF7, AF8, TP9, TP10) and a reference at Fpz. It supports a maximum sampling rate of 256 Hz. Signal acquisition was performed using BlueMuse, an unofficial external application that streams data via the Lab Streaming Layer (LSL) protocol for integration with OpenViBE.

The DSI-24 (Wearable Sensing, Inc.) served as the research-grade reference device for comparison with the four consumer-grade systems. It features 21 electrodes, supports both wired and wireless connections, and offers a maximum sampling rate of 300 Hz. The DSI-24 has been widely adopted in studies involving P300 spellers, motor imagery, and quantitative EEG (QEEG), consistently demonstrating reliable performance.

EEG data pre-processing. All uploaded datasets are provided without any preprocessing. The datasets include recordings from all available channels; however, for the analysis, we used the following left frontal channel for each device: Fp1 for BLP, MW2, and DSI-24; AF7 for Muse 2; and EEG1 for FX2.

A list of datasets excluded from analysis due to excessive noise is provided. These recordings are not included in the publicly available dataset.

Eye blinking and jaw clenching

Muse2: sub-05, sub-06, sub-07
FX2: sub-24

Head movement

Brain Link Pro: sub-26

```
EEG
├── sourcedata
│   ├── sub-01
│   │   ├── sub-01_task-BT_acq-BLP_eeg.edf
│   │   ├── sub-01_task-BT_acq-DSI_eeg.edf
│   │   ├── sub-01_task-BT_acq-FX2_eeg.txt
│   │   ├── sub-01_task-BT_acq-Muse2_eeg.edf
│   │   ├── sub-01_task-BT_acq-MW2_eeg.edf
│   │   ├── sub-01_task-EB_acq-BLP_eeg.edf
│   │   ├── sub-01_task-EB_acq-DSI_eeg.edf
│   │   ├── sub-01_task-EB_acq-FX2_eeg.txt
│   │   ├── sub-01_task-EB_acq-Muse2_eeg.edf
│   │   ├── sub-01_task-EB_acq-MW2_eeg.edf
│   │   ├── sub-01_task-MVC_acq-BLP_eeg.edf
│   │   ├── sub-01_task-MVC_acq-DSI_eeg.edf
│   │   ├── sub-01_task-MVC_acq-FX2_eeg.txt
│   │   ├── sub-01_task-MVC_acq-Muse2_eeg.edf
│   │   ├── sub-01_task-MVC_acq-MW2_eeg.edf
│   │   ├── sub-01_task-MVO_acq-BLP_eeg.edf
│   │   ├── sub-01_task-MVO_acq-DSI_eeg.edf
│   │   ├── sub-01_task-MVO_acq-FX2_eeg.txt
│   │   ├── sub-01_task-MVO_acq-Muse2_eeg.edf
│   │   └── sub-01_task-MVO_acq-MW2_eeg.edf
│   ├── ...
│   └── sub-30
├── derivatives
│   ├── sub-01
│   │   ├── sub-01_task-BT_acq-BLP_eeg.mat
│   │   ├── sub-01_task-BT_acq-DSI_eeg.mat
│   │   ├── sub-01_task-BT_acq-FX2_eeg.mat
│   │   ├── sub-01_task-BT_acq-Muse2_eeg.mat
│   │   ├── sub-01_task-BT_acq-MW2_eeg.mat
│   │   ├── sub-01_task-EB_acq-BLP_eeg.mat
│   │   ├── sub-01_task-EB_acq-DSI_eeg.mat
│   │   ├── sub-01_task-EB_acq-FX2_eeg.mat
│   │   ├── sub-01_task-EB_acq-Muse2_eeg.mat
│   │   ├── sub-01_task-EB_acq-MW2_eeg.mat
│   │   ├── sub-01_task-MVC_acq-BLP_eeg.mat
│   │   ├── sub-01_task-MVC_acq-DSI_eeg.mat
│   │   ├── sub-01_task-MVC_acq-FX2_eeg.mat
│   │   ├── sub-01_task-MVC_acq-Muse2_eeg.mat
│   │   ├── sub-01_task-MVC_acq-MW2_eeg.mat
│   │   ├── sub-01_task-MVO_acq-BLP_eeg.mat
│   │   ├── sub-01_task-MVO_acq-DSI_eeg.mat
│   │   ├── sub-01_task-MVO_acq-FX2_eeg.mat
│   │   ├── sub-01_task-MVO_acq-Muse2_eeg.mat
│   │   └── sub-01_task-MVO_acq-MW2_eeg.mat
│   ├── ...
│   └── sub-30
├── CHANGES
├── dataset_description.json
├── participants.json
├── participants.tsv
├── example_script.m
├── example_sctipt.py
├── survey_result.csv
└── README
```

Fig. 2 File structure.

FX2: sub-26
Mindwave2: sub-26
Muse2: sub-05, sub-06, sub-07, sub-16, sub-26
DSI-24: sub-26

Data Records

The dataset is publicly available on Figshare[12]. As noted above, 30 participants completed four sessions with each of the five EEG devices. Under the sourcedata directory, subfolders are organized by participant, and all files within each participant's folder contain raw data. Survey responses on device usability are provided in surve (Fig. 2).

Code	Label	Meaning	Operational definition
1	PreRest_Start	Experiment start (pre-rest)	Start of the pre-rest baseline period.
2	Task_Ready	Pre-rest end; preparation beep	End of pre-rest; preparation cue (beep) is presented.
3	Task_Onset	Task	Onset of each task repetition (one marker per task repetition).
4	PostRest_Start	Task end; post-rest start	End of the task period; start of the post-rest period.
5	Experiment_End	Experiment end	End of the experimental session.

Table 3. Event codebook (shared across EDF and MAT).

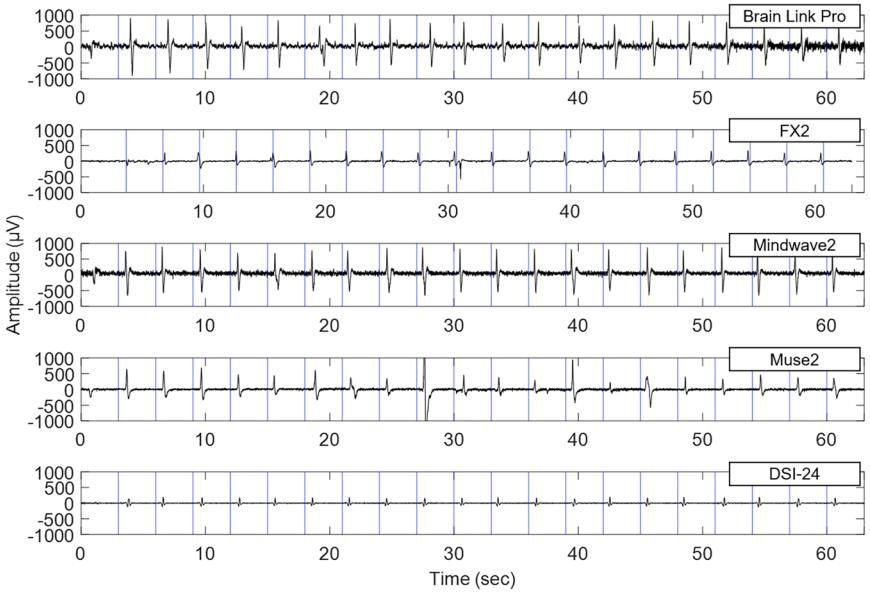


Fig. 3 Raw EEG signals during eye-blink task (Subject 04).

Event information is provided in both the EDF and MAT files to support reproducible segmentation of the recording into pre-rest, task, and post-rest periods. In the EDF files, event codes are stored under the event structure, with the integer code written in the edftype field (values 1–5). In the MAT files, the corresponding event information is stored in the top-level variable under the.event field, using the same integer coding scheme as the EDF files. Table 3 summarizes the event codebook used in this dataset; notably, code 3 indicates the onset of each task repetition (i.e., one marker per repetition).

Technical Validation

Technical validation covers physical signal detection, brain wave detection, and robustness to movement-related noise.

For detecting physical signals that were induced by the instructed movement, EEG recordings were visually inspected to confirm their presence in the raw data. Figure 3 shows plots from all five devices for participant s04, illustrating the detection of such physical signals.

For brain wave detection, we examined whether characteristic EEG features were observed. Specifically, we confirmed the well-established phenomenon whereby alpha-band (8–13 Hz) power increases when the eyes are closed compared with when they are open, with a clear spectral peak in the alpha-band during eyes-closed conditions. Figure 4 presents frequency analyses from all five devices for participant s04, demonstrating this characteristic response.

For robustness to movement-related noise, frequency spectra before and after movement were compared using correlation analysis. Figure 5 shows the results of this comparison for participant s04 across the five devices.

Signal detection. Figure 3 shows signals recorded during eye-blink tasks, plotted for signal detection. The vertical blue lines indicate the timing of beep cues, and corresponding peaks can be observed at these time points. To detect eye-blink and jaw-clenching events, raw signals were visualized using plots similar to Fig. 3. Three independent raters manually identified the events, and the results were cross-checked for consistency. Eye blinks were consistently detected 20 times across all five devices for all participants. For jaw clenching, the mean number of detected events across participants was 20 for BLP, 19.90 for FX2, 20 for MW2, 19.85 for Muse 2, and 20 for DSI-24.

Brain wave detection. Figure 4 shows the frequency spectrum obtained during eyes-open and eyes-closed conditions. The blue line represents the spectrum for eyes open, whereas the red line represents the spectrum for

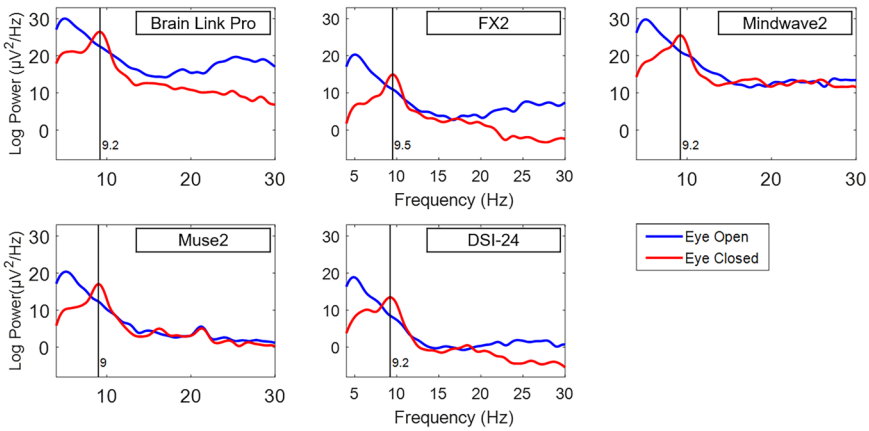


Fig. 4 Power spectral density of eye-open vs. eye-closed conditions (Subject 04).

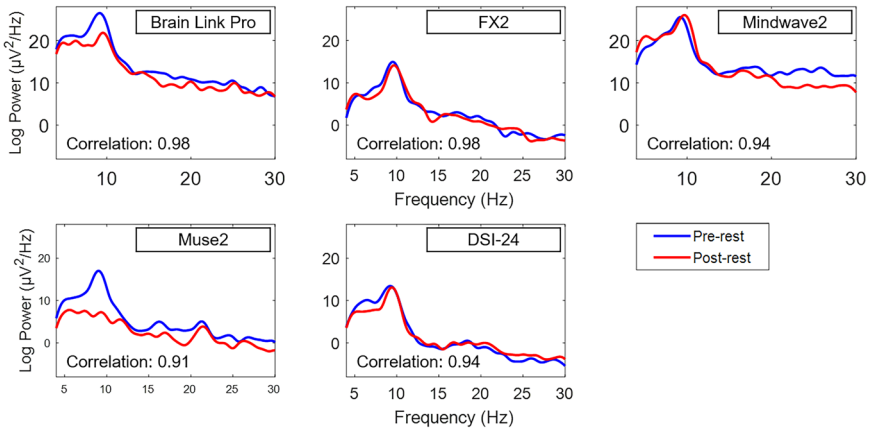


Fig. 5 Power spectral density before and after movement (Subject 04).

eyes closed. As expected, the alpha-band (8–13 Hz) power increased in the eyes-closed condition compared with the eyes-open condition, with a distinct alpha peak observed during eyes closed.

Although the alpha peak frequency varies across individuals, it tends to remain consistent for a given participant regardless of the recording device. Using the research-grade DSI-24 as a reference, we verified that the consumer-grade devices captured alpha peaks at comparable frequencies. As shown in Fig. 4, all five devices demonstrated alpha activity within a similar frequency range.

Across all participants, the mean absolute differences in the alpha peak frequency between the research-grade device and each consumer-grade device were $\Delta f = 0.24$ Hz (BLP), 0.26 Hz (FX2), 0.20 Hz (MW2), and 0.32 Hz (Muse 2). No statistically significant differences were observed ($p > 0.05$).

Noise robustness detection. Figure 5 shows frequency spectra obtained before and after head-movement tasks. The blue line represents the spectrum before movement, and the red line represents the spectrum after movement. Both conditions were recorded with eyes closed. If the difference between pre- and post-movement signals is minimal, the spectra are expected to be highly correlated; otherwise, the correlation would be reduced. To quantify this, correlation coefficients were calculated. For the spectra shown in Fig. 5, the correlation coefficients were 0.98 (BLP), 0.98 (FX2), 0.94 (MW2), 0.91 (Muse 2), and 0.99 (DSI-24). Across all participants, the mean correlation coefficients were 0.95 (BLP), 0.91 (FX2), 0.94 (MW2), 0.89 (Muse 2), and 0.94 (DSI-24).

Code availability

The code for basic data visualization and spectral analysis is available alongside the data in the same public repository. These codes allow users to reproduce figures similar to those presented in this article.

Data availability

The dataset supporting this article is available at figshare: https://doi.org/10.6084/m9.figshare.30162868.

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Author contributions

Data acquisition: Y.L., D.G., K.K., S.P., S.S., M.U.C. Data curation: Y.L., D.G., K.K., S.P., S.S., M.U.C. Validation: Y.L., D.G., M.A. Formal analysis: Y.L. Investigation: Y.L., D.G., M.A. Visualization: Y.L., D.G., M.A. Writing: Y.L., D.G., M.A.

Competing interests

The authors declare no competing interests.

Additional information

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